

FACTSHEET FS09 · 52-week high

What changed on 2 October 2026. Two corrections, both to the universe of stocks, and every number below is re-run on them. (1) **US:** the universe is now the actual S&P 500 members at each month-end. Until now it was the 500 largest of every company that was *ever* in the index 2012–2026, which includes later winners before they joined. (2) **UK and EU:** the price panels now include the index members that stopped trading (UK coverage of members 71% → 92%, EU 81% → 89%); before, most of them had no prices. SCANDI stays as registered (OMXC25 + OMXS30 + OMXH25). Registered verdicts are unchanged; where a corrected number crosses a gate, the text says so. Comparison and method: FACTSHEETS_RERUN_2026-10-02.md.

Do stocks near their high keep going?

Henrik Gjerning · Rude Investment Consulting · Project 10 sidebar: strategy factsheets · data to 31 August 2026

STRATEGY	ORIGIN	FAMILY
Anchoring momentum	George & Hwang (2004)	Momentum & trend
UNIVERSES	BACKTEST	BOOKS
US · EU · UK · DK · SCANDI · World (point-in-time)	Feb 2013 – Aug 2026, monthly, net of costs	Long/short and long-only (stocks closest to their 52-week high)
REBALANCE	STATUS	
Monthly	Descriptive factsheet, not a registered trial	

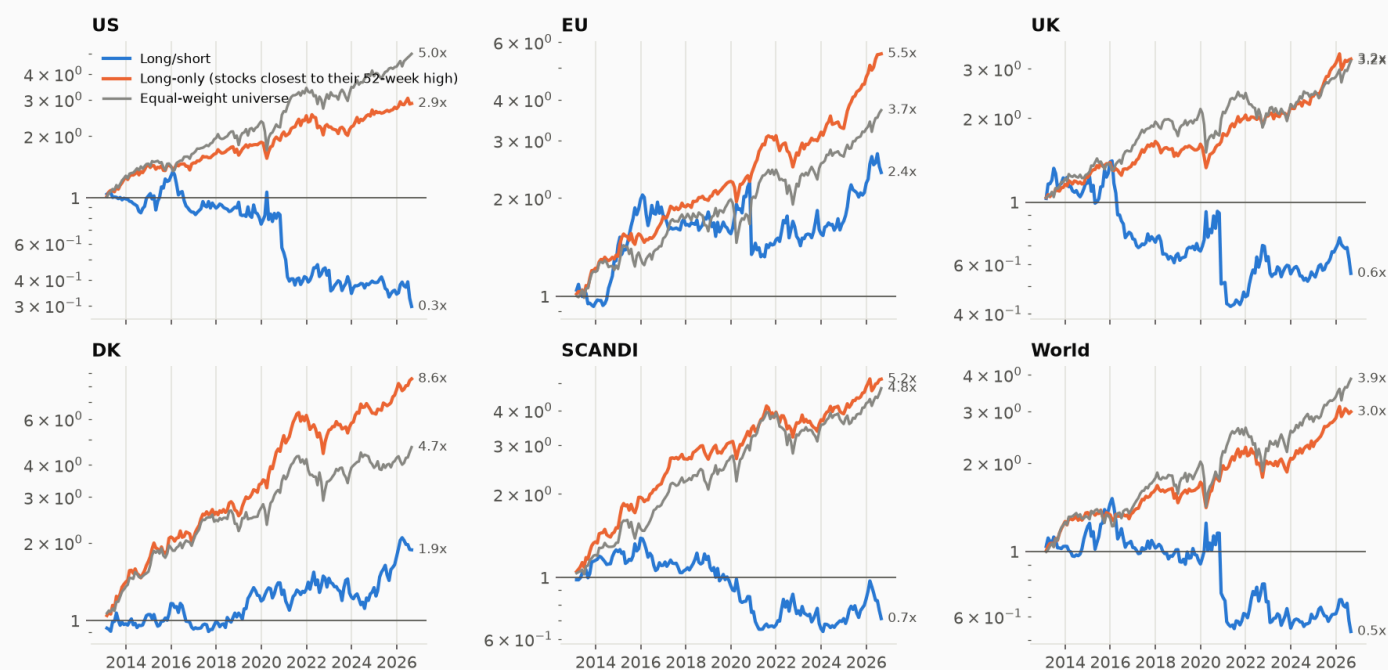
The claim. Stocks trading close to their 52-week high outperform those far below it, and this nearness explains much of momentum. George & Hwang (2004) found it for US stocks 1963–2001; investors anchor on the high and under-react to good news near it. **What we find, 2013–2026, point-in-time universes, after costs:** the long/short earned -5.8% to +8.8% a year and was positive in 2 of the six; no universe passes the 2.87 gate. The long-only book (stocks closest to their 52-week high) had a higher Sharpe than the equal-weight universe in 5 of the six (EU, UK, DK, SCANDI, World), with alphas of +0.1% to +6.9%, passing the gate in EU, DK. **What goes wrong (§10):** the main drags are a market bet (average beta -0.66, worth -8.1% a year in 2013–19), costs of about 2.0% a year. None of the pre-declared fixes passes the gate in both windows. **Classification (§11):** Momentum & trend; its returns sit closest to the momentum cluster of the signal library. **360° view (§12):** nearest neighbours 55-day breakout, MIN5; after its closest neighbours and the market it keeps an alpha of -1.2% to +3.8% (largest |t| 2.6), so it carries some information of its own.

1. Headline performance

Universe	Names / groups	L/S return / yr	L/S t	L/S CAPM alpha (t)	L/S beta	Long-only CAGR	Long-only Sharpe	Long-only max DD	EW CAGR	EW Sharpe	Size-weighted CAGR
US	502 / 10	-5.8%	-1.00	+6.8% (1.7)	-0.97	+8.1%	0.70	-21%	+12.6%	0.85	+14.9%
EU	264 / 10	+8.8%	1.78	+18.6% (4.5)	-0.91	+13.4%	1.25	-20%	+10.1%	0.75	+10.2%
UK	327 / 10	-0.9%	-0.13	+9.5% (1.9)	-1.09	+9.1%	0.90	-20%	+8.9%	0.68	+7.8%
DK	20 / 3	+6.1%	1.56	+11.0% (3.0)	-0.39	+17.1%	1.16	-31%	+12.0%	0.83	+10.2%
SCANDI	70 / 5	-1.2%	-0.29	+6.3% (1.9)	-0.59	+12.9%	1.04	-23%	+12.2%	0.89	+11.0%
World	1089 / 10	-1.9%	-0.34	+8.6% (2.3)	-0.93	+8.4%	0.74	-21%	+10.5%	0.71	+10.3%

Monthly, local currency (World in USD). L/S = equal-weighted stocks closest to their 52-week high minus stocks furthest below it, net of 10 bp per side on turnover, a size-tiered borrow fee on the short leg and, where the book is net long cash, financing at the USD risk-free rate + 0.5%. Newey-West t (6 lags). CAPM vs the equal-weight universe. Size-weighted universe = market-cap weights in the US, traded-value weights elsewhere (a proxy for the index).

52-week high on six stock universes, 2013-2026



Growth of 1, log scale, local currency, net of costs. Henrik Gjering · Project 10 factsheets

2. Strategy description

Stocks trading close to their 52-week high outperform those far below it, and this nearness explains much of momentum. George & Hwang (2004) found it for US stocks 1963–2001; investors anchor on the high and under-react to good news near it.

Why it might work. Anchoring: traders use the 52-week high as a reference and are reluctant to bid prices above it, so good news is absorbed slowly (George & Hwang 2004).

Rule	Original (George & Hwang (2004))	This factsheet
Signal	Price / 52-week high	Total-return price / 252-day high at month-end t
Portfolio	Top 30% minus bottom 30%, 6-month holding	Extreme quantile L/S, one-month holding
Weighting	Equal weighted	Equal weighted

3. Data load

Universe	Prices	Membership	Names eligible (median)	Currency
US	Sharadar SEP ccloseadj (total return), 805 tickers incl. delisted	Point-in-time: the S&P 500 members at each month-end (Sharadar add/remove history; corrected 2 Oct 2026)	~503	USD
EU	Project1 yfinance cache, Adj Close (TR)	Point-in-time: union of 11 national blue-chip indices (CAC 40, DAX, AEX, IBEX 35, FTSE MIB, OMXS30, OMXC25, OMXH25, BEL 20, PSI-20, WIG20), quarterly Wikipedia snapshots 2012–2026; 83% of member-quarters priced	~238	local (EUR, SEK, DKK, PLN)
UK	Project1 cache, Close + pence dividends (Yahoo under-adjusts LSE ~100×)	Point-in-time: FTSE 100 + FTSE 250, quarterly snapshots; 75% priced	~242	GBP
DK	Project1 cache, Adj Close	Point-in-time: OMX Copenhagen 25, quarterly snapshots; 91% priced	~19	DKK
SCANDI	Project1 cache, Adj Close	Point-in-time: OMXC25 + OMXS30 + OMXH25 (Oslo has no snapshot history); 88% priced	~62	local (DKK, SEK, EUR)
World	US Sharadar + UK + EU above	Point-in-time union of the three; local-currency returns summed, not converted	~963	mixed

Membership comes from Project2's cache of dated Wikipedia index pages (the same parsing as run_wiki_pit.py / run_eu_pit.py), with a short list of verified ticker renames (e.g. NZYM-B → NSIS-B, WDH → DEMANT, DAI → MBG). A name is eligible from the day after the snapshot that lists it, once it has 60 days of prices. Member-quarters without a price are mostly delisted names that Yahoo no longer serves, so some survivorship remains; see §13.

- **Data gap:** We hold for one month; George & Hwang hold for six. A longer holding period would cut turnover.

4. Signal creation

$HI_{52,i,t} = P_{i,t} / \max(P_{i,t-251..t})$. Signals use only information up to month-end t and are traded in month $t+1$. Eligibility: in the point-in-time universe on any of the last five trading days of month t .

5. Model build

- Sort eligible stocks into equal-count groups (deciles ≥ 100 names, quintiles 50–99, terciles < 50 ; fixed per universe).
- **Long/short** = stocks closest to their 52-week high minus stocks furthest below it; **long-only** = stocks closest to their 52-week high; benchmarks: equal-weight universe and size-weighted universe.
- Equal weights within each leg, rebalanced monthly.
- Costs: 10 bp per side on the one-way turnover of each leg; short-leg borrow fee 0.25% / 0.75% / 2% a year by size tier (largest 50% / next 30% / smallest 20% of the universe, `code/borrow.py`); net long cash financed at the USD risk-free rate + 0.5%. Code: `code/xs.py` (engine), `code/analyse_xs.py`, `code/build_xs.py`.

6. Performance in detail

US (S&P 500 members, point-in-time), median 502 names, 10 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	-5.8%	-8.5%	24.3%	-0.24 [-0.69, 0.20]	-0.45	-78%	50%	-1.00	-0.97	+6.8% (1.7)	18.2%
Long-only (net)	+8.5%	+8.1%	12.3%	0.70 [0.28, 1.16]	1.03	-21%	61%	3.64	0.65	+0.1% (0.0)	7.4%
Stocks closest to their 52-week high (gross)	+10.3%	+9.9%	12.3%	0.84 [0.41, 1.31]	1.31	-20%	63%	4.34	0.65	+1.8% (1.2)	7.3%
Stocks furthest below it (gross)	+12.6%	+8.8%	28.5%	0.44 [0.00, 0.96]	0.50	-50%	56%	1.89	1.61	-8.6% (-2.8)	15.9%
EW universe	+13.1%	+12.6%	15.5%	0.85 [0.41, 1.42]	1.29	-28%	67%	4.20	1.00	–	9.2%
Size-weighted universe	+14.9%	+14.9%	14.2%	1.05 [0.60, 1.61]	1.73	-24%	69%	5.29	0.86	+3.7% (2.4)	8.3%

EU (11 national blue-chip indices, point-in-time), median 264 names, 10 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	+8.8%	+6.6%	20.9%	0.42 [-0.03, 0.99]	0.43	-41%	55%	1.78	-0.91	+18.6% (4.5)	13.6%
Long-only (net)	+13.2%	+13.4%	10.5%	1.25 [0.70, 1.90]	2.31	-20%	65%	4.57	0.59	+6.9% (4.1)	5.7%
Stocks closest to their 52-week high (gross)	+14.8%	+15.2%	10.5%	1.40 [0.84, 2.06]	2.70	-19%	66%	5.09	0.59	+8.4% (5.0)	5.5%
Stocks furthest below it (gross)	+2.9%	-0.0%	24.8%	0.12 [-0.37, 0.57]	-0.00	-54%	49%	0.47	1.50	-13.3% (-4.0)	12.6%
EW universe	+10.7%	+10.1%	14.3%	0.75 [0.29, 1.30]	1.14	-26%	62%	3.18	1.00	–	8.4%
Size-weighted universe	+10.6%	+10.2%	13.7%	0.77 [0.32, 1.29]	1.21	-25%	58%	3.37	0.92	+0.8% (0.6)	7.9%

UK (FTSE 350, point-in-time), median 327 names, 10 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	-0.9%	-4.2%	24.9%	-0.04 [-0.51, 0.48]	-0.22	-70%	55%	-0.13	-1.09	+9.5% (1.9)	18.0%
Long-only (net)	+9.2%	+9.1%	10.3%	0.90 [0.42, 1.45]	1.39	-20%	60%	3.74	0.57	+3.7% (1.9)	6.5%
Stocks closest to their 52-week high (gross)	+10.9%	+10.8%	10.3%	1.06 [0.56, 1.62]	1.71	-17%	62%	4.38	0.57	+5.3% (2.8)	6.4%
Stocks furthest below it (gross)	+8.9%	+5.2%	28.0%	0.32 [-0.15, 0.80]	0.31	-50%	52%	1.22	1.66	-7.0% (-1.8)	15.7%
EW universe	+9.6%	+8.9%	14.2%	0.68 [0.20, 1.28]	0.96	-30%	58%	2.87	1.00	–	8.8%
Size-weighted universe	+8.4%	+7.8%	12.9%	0.65 [0.16, 1.25]	0.90	-28%	62%	2.68	0.84	+0.3% (0.2)	7.8%

DK (OMXC25, point-in-time), median 20 names, 3 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	+6.1%	+4.7%	17.1%	0.36 [-0.08, 0.82]	0.43	-28%	53%	1.56	-0.39	+11.0% (3.0)	9.3%
Long-only (net)	+17.0%	+17.1%	14.6%	1.16 [0.63, 1.70]	2.18	-31%	63%	4.40	0.81	+6.8% (3.8)	7.0%
Stocks closest to their 52-week high (gross)	+17.7%	+18.0%	14.7%	1.21 [0.67, 1.76]	2.31	-30%	63%	4.57	0.81	+7.5% (4.2)	7.0%
Stocks furthest below it (gross)	+9.6%	+7.7%	20.9%	0.46 [-0.06, 1.07]	0.55	-44%	59%	1.72	1.21	-5.5% (-2.2)	12.4%
EW universe	+12.6%	+12.0%	15.2%	0.83 [0.31, 1.44]	1.27	-33%	66%	3.31	1.00	–	8.8%
Size-weighted universe	+11.1%	+10.2%	16.4%	0.68 [0.11, 1.33]	0.94	-40%	62%	2.55	0.93	-0.6% (-0.2)	10.7%

SCANDI (OMXC25 + OMXS30 + OMXH25, point-in-time), median 70 names, 5 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	-1.2%	-2.5%	16.2%	-0.07 [-0.54, 0.40]	-0.21	-54%	52%	-0.29	-0.59	+6.3% (1.9)	9.6%
Long-only (net)	+12.9%	+12.9%	12.4%	1.04 [0.52, 1.61]	1.77	-23%	66%	4.21	0.76	+3.4% (2.3)	6.6%
Stocks closest to their 52-week high (gross)	+14.1%	+14.1%	12.4%	1.13 [0.61, 1.70]	1.98	-22%	66%	4.55	0.76	+4.5% (3.0)	6.5%
Stocks furthest below it (gross)	+12.7%	+11.0%	21.4%	0.59 [0.12, 1.13]	0.82	-44%	59%	2.42	1.35	-4.3% (-1.8)	12.6%
EW universe	+12.6%	+12.2%	14.2%	0.89 [0.37, 1.52]	1.38	-29%	66%	3.63	1.00	–	8.4%
Size-weighted universe	+11.4%	+11.0%	13.5%	0.84 [0.37, 1.41]	1.32	-26%	63%	3.59	0.90	-0.0% (-0.0)	8.1%

World (US + UK + EU, point-in-time), median 1089 names, 10 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	-1.9%	-4.5%	22.3%	-0.09 [-0.54, 0.39]	-0.26	-65%	53%	-0.34	-0.93	+8.6% (2.3)	16.7%

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
Long-only (net)	+8.8%	+8.4%	12.0%	0.74 [0.29, 1.28]	1.09	-21%	63%	3.39	0.64	+1.6% (1.1)	7.6%
Stocks closest to their 52-week high (gross)	+10.5%	+10.2%	12.0%	0.88 [0.41, 1.44]	1.37	-20%	63%	4.01	0.64	+3.2% (2.4)	7.5%
Stocks furthest below it (gross)	+9.2%	+5.6%	27.8%	0.33 [-0.13, 0.81]	0.33	-47%	54%	1.32	1.58	-8.6% (-3.2)	15.0%
EW universe	+11.3%	+10.5%	16.0%	0.71 [0.25, 1.28]	1.02	-30%	63%	3.07	1.00	–	9.7%
Size-weighted universe	+11.0%	+10.3%	14.8%	0.74 [0.28, 1.31]	1.10	-28%	61%	3.19	0.91	+0.7% (0.8)	8.9%

Sharpe CI: stationary bootstrap (mean block 6 months, 5,000 draws). Max drawdown on monthly returns.

Calendar-year returns, L/S (net)

Year	US	EU	UK	DK	SCANDI	World
2013	-1.3%	-4.4%	+20.8%	-2.9%	+15.4%	+4.2%
2014	+3.6%	+49.8%	-3.1%	+8.7%	+7.4%	+16.7%
2015	+24.7%	+38.3%	+19.3%	+10.2%	+11.8%	+20.0%
2016	-33.2%	-10.7%	-47.0%	-20.4%	-20.2%	-29.3%
2017	+4.6%	-6.5%	-2.7%	+7.1%	-3.5%	-0.4%
2018	+6.9%	+9.7%	-8.9%	+7.6%	+11.8%	-1.2%
2019	-21.8%	-14.2%	+2.0%	+11.3%	-21.8%	-10.7%
2020	-27.3%	-11.1%	-23.1%	+10.2%	-19.1%	-23.1%
2021	-24.4%	+7.8%	-10.7%	+0.9%	-10.6%	-12.4%
2022	+8.6%	+0.3%	+39.2%	+10.2%	+12.5%	+10.6%
2023	-22.9%	+3.9%	-13.3%	-15.5%	-14.0%	-18.9%
2024	+13.2%	+9.3%	-0.7%	+8.8%	+9.0%	+8.2%
2025	-12.3%	+30.7%	+24.5%	+31.8%	+18.0%	+3.4%
2026	-12.8%	+7.8%	-18.8%	+6.2%	-15.3%	-13.0%

Calendar-year returns, long-only (net)

Year	US	EU	UK	DK	SCANDI	World
2013	+26.2%	+21.9%	+15.8%	+41.0%	+34.5%	+26.5%
2014	+14.5%	+8.8%	+5.9%	+20.0%	+12.1%	+7.3%
2015	-1.8%	+16.0%	+13.0%	+24.3%	+29.1%	-1.8%
2016	+4.2%	+14.0%	-0.2%	-0.8%	+16.9%	+1.7%

Year	US	EU	UK	DK	SCANDI	World
2017	+11.1%	+7.1%	+19.3%	+25.6%	+17.5%	+19.8%
2018	-1.9%	+4.7%	-11.3%	-3.1%	-0.3%	-9.9%
2019	+15.6%	+14.4%	+10.2%	+33.0%	+15.3%	+17.6%
2020	+12.2%	+9.5%	+9.0%	+53.5%	+14.4%	+13.8%
2021	+19.9%	+26.0%	+16.6%	+19.7%	+12.1%	+15.6%
2022	-11.9%	-12.8%	-0.9%	-12.3%	-8.7%	-12.3%
2023	+0.3%	+17.6%	+7.7%	+6.5%	+2.9%	+4.5%
2024	+15.0%	+2.6%	+9.2%	+9.2%	+5.0%	+9.6%
2025	+6.6%	+43.7%	+31.3%	+23.3%	+25.1%	+25.0%
2026	+5.9%	+17.8%	+3.5%	+10.1%	+6.6%	+5.1%

7. Trading record

The rebalance log per universe is in `results/xs_FS09_record_<universe>.csv` (names per leg, turnover, leg returns, gross and net L/S).

Universe	Rebalances	Names long / short	Turnover long leg / month	Turnover short leg / month	L/S months positive	Worst L/S month	When
US	163	51 / 50	71%	24%	50%	-30.9%	2020-11
EU	163	27 / 26	65%	22%	55%	-39.8%	2020-11
UK	163	33 / 32	68%	24%	55%	-44.2%	2020-11
DK	163	7 / 6	29%	21%	53%	-11.8%	2023-11
SCANDI	163	14 / 14	47%	24%	52%	-12.3%	2020-11
World	163	109 / 108	70%	23%	53%	-37.6%	2020-11

Current book (signal at 31 August 2026; first 8 names per leg)

Universe	Long: stocks closest to their 52-week high	Short: stocks furthest below it
US	VLO, PSX, SLB, CRWD, MPC, GEN, TECH, PFE	TTD, CSGP, FISV, PODD, APP, BLDR, BSX, ORCL
EU	ASRNL.AS, RAND.AS, SKF-B.ST, ABN.AS, DSFIR.AS, JYSK.CO, KESKOB.HE, PUIG.MC	ONTEX.BR, FCT.MI, AVIO.MI, STLAM.MI, AGFB.BR, STLAP.PA, NYR.BR, UMG.AS
UK	ASL.L, RS1.L, ATYM.L, HAS.L, GCP.L, AAL.L, VCT.L, INVP.L	AML.L, VTY.L, CWR.L, TEP.L, RPI.L, DATA.L, OXB.L, SMWH.L

Universe	Long: stocks closest to their 52-week high	Short: stocks furthest below it
DK	JYSK.CO, NDA-DK.CO, NSIS-B.CO, PNDORA.CO, VWS.CO, DANSKE.CO, ISS.CO, DEMANT.CO	ZEAL.CO, AMBU-B.CO, ORSTED.CO, RBREW.CO, DSV.CO, COLO-B.CO, NOVO-B.CO, GN.CO
SCANDI	KESKOB.HE, JYSK.CO, SKF-B.ST, NDA-SE.ST, NDA-DK.CO, NSIS-B.CO, NDA-FI.HE, SHB-A.ST	NOKIA.HE, ORSTED.CO, RBREW.CO, AMBU-B.CO, QTCOM.HE, WRT1V.HE, ERIC-B.ST, COLO-B.CO

8. Risk and factor attribution

JKP 7 themes (US, UK, DK, World) or French Europe 5F + WML (EU, SCANDI); Newey-West t, 6 lags.

Long/short

Universe	Factor model	Alpha / yr	t(alpha)	R²	Loadings with t ≥ 2 (t)
US	JKP USA 7 themes	-9.3%	-4.49	0.87	value -0.70 (-5.4), momentum +1.09 (+7.4), low_risk +1.29 (+6.7), quality +0.85 (+3.2), short_term_reversal -1.07 (-4.9)
EU	French Europe 5F + WML	-0.7%	-0.19	0.62	Mkt-RF -0.41 (-4.5), WML +1.19 (+6.6)
UK	JKP GBR 7 themes	-6.6%	-1.49	0.78	momentum +1.09 (+6.9), low_risk +1.66 (+5.7), quality +0.75 (+2.4), short_term_reversal -0.94 (-4.6)
DK	JKP DNK 7 themes	-3.0%	-1.20	0.60	momentum +1.00 (+10.7), low_risk +0.65 (+5.7)
SCANDI	French Europe 5F + WML	-9.0%	-2.93	0.45	Mkt-RF -0.25 (-3.0), HML +0.46 (+2.4), WML +0.87 (+9.0)
World	JKP World 7 themes	-2.5%	-0.57	0.72	mkt -0.16 (-2.1), value -0.51 (-3.1), momentum +0.97 (+7.4), low_risk +1.02 (+5.5), short_term_reversal -0.99 (-4.1)

Long-only

Universe	Factor model	Alpha / yr	t(alpha)	R²	Loadings with t ≥ 2 (t)
US	JKP USA 7 themes	-2.4%	-1.79	0.86	mkt +0.92 (+26.1), momentum +0.32 (+3.6), low_risk +0.48 (+6.0), short_term_reversal -0.36 (-4.8)
EU	French Europe 5F + WML	+5.8%	2.73	0.55	Mkt-RF +0.51 (+10.8), HML +0.30 (+2.7), WML +0.25 (+4.4)
UK	JKP GBR 7 themes	+3.4%	1.80	0.57	mkt +0.42 (+9.7), value +0.19 (+2.6), momentum +0.31 (+4.2), quality +0.46 (+4.5)
DK	JKP DNK 7 themes	+6.4%	2.52	0.64	mkt +0.73 (+16.2), momentum +0.26 (+3.6), quality +0.20 (+2.5), short_term_reversal -0.19 (-2.4)
SCANDI	French Europe 5F + WML	+5.5%	2.15	0.49	Mkt-RF +0.58 (+9.9), RMW +0.37 (+2.4), WML +0.21 (+2.7)
World	JKP World 7 themes	+3.2%	1.32	0.67	mkt +0.66 (+13.2), momentum +0.27 (+3.1), low_risk +0.35 (+3.0), short_term_reversal -0.33 (-2.2)

9. Statistical verdict

- **Gate:** $|t| > 2.87$ (2 books $\times$ 6 universes, Bonferroni $0.05/12$, two-sided).
- **L/S mean:** positive in 2 of the six; passes in 0 of 6.
- **L/S CAPM alpha:** +6.3% to +18.6%, $|t|$ past the gate in 2 of 6.
- **Long-only:** Sharpe above the equal-weight universe in 5 of the six; alpha +0.1% to +6.9%, passing in 2 of 6.
- **Probabilistic Sharpe** of the L/S, $P(\text{true Sharpe} > 0)$: US 0.19, EU 0.91, UK 0.45, DK 0.90, SCANDI 0.39, World 0.37.
- **Publication decay:** gross L/S -0.9% to +11.0% a year in 2013–19 against -5.8% to +12.8% in 2020–26.

10. What goes wrong, and how to fix it

Method. Diagnose on the design window (2013–2019); apply the same pre-declared fix ladder used for every sort strategy in FS03–FS12; judge on the holdout (Jan 2020 – Aug 2026) with a pooled six-universe paired bootstrap of the Sharpe difference against the previous step. 25 fix trials across FS03–FS12, gate $p < 0.0020$.

10.1 Diagnosis (design window 2013–2019)

Universe	L/S gross / yr	Beta	Beta drag / yr	CAPM alpha (gross)	Costs / yr	Long leg vs EW / yr	EW vs short leg / yr	Turnover / month	Worst months
US	+1.1%	-0.83	-10.9%	+12.0%	-2.3%	-1.9%	+3.0%	97%	2015-04 -15%, 2019-01 -13%, 2016-04 -11%
EU	+11.0%	-0.91	-9.7%	+20.7%	-2.0%	+3.1%	+7.9%	85%	2015-02 -12%, 2016-03 -11%, 2019-04 -9%
UK	-0.9%	-0.75	-8.7%	+7.8%	-2.2%	-2.7%	+1.7%	93%	2016-02 -17%, 2016-04 -17%, 2015-04 -14%
DK	+5.6%	-0.16	-2.4%	+8.0%	-1.4%	+4.0%	+1.6%	57%	2016-08 -7%, 2016-07 -7%, 2013-02 -7%
SCANDI	+2.8%	-0.52	-8.0%	+10.7%	-1.8%	+2.9%	-0.1%	75%	2019-06 -10%, 2015-04 -9%, 2019-01 -9%
World	+3.3%	-0.81	-8.9%	+12.2%	-2.3%	-0.9%	+4.2%	94%	2016-04 -14%, 2015-04 -13%, 2015-02 -10%

On average across the six universes the gross spread was +3.8% a year, of which the market exposure (beta -0.66) contributed -8.1%; the beta-adjusted spread (CAPM alpha) was +11.9%. Costs took 2.0% a year at 83% monthly turnover across both legs. The long leg beat the universe by +0.8% and the short leg lagged it by +3.1% a year, so

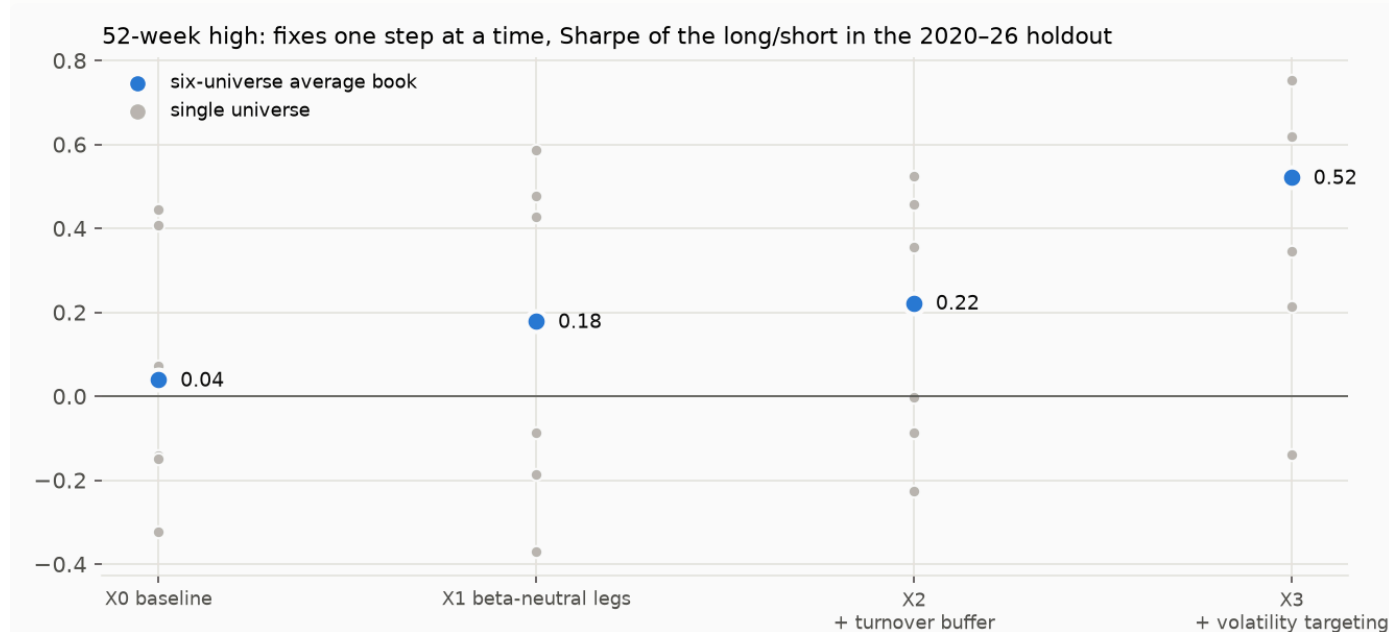
most of the spread comes from the short side. Holding longer does not change the picture much: the gross spread 2, 3 and 6 months after formation averages +3.5%, +4.0%, +0.3% a year.

10.2 The fix ladder

X0 baseline → X1 **beta-neutral legs** (each leg scaled by 1/its ex-ante beta) → X2 + **turnover buffer** (enter the extreme 1/q, keep a name until it leaves the extreme 3/q) → X3 + **volatility targeting** (scale the L/S to 10% a year using its trailing 6-month volatility, lagged; Barroso & Santa-Clara 2015; Moreira & Muir 2017).

Step	Design Sharpe (avg)	Δ design (p)	Holdout Sharpe (book)	Δ holdout (p)	Universes improved	Holdout return / yr	Holdout max DD	Holdout alpha vs EW (t)	Costs / yr
X0 baseline	0.06	–	0.04	–	–	+0.8%	-42%	+10.4% (2.5)	2.0%
X1 beta-neutral legs	0.37	+0.40 (p 0.001)	0.18	+0.14 (p 0.179)	4 of 6	+2.6%	-31%	+5.9% (1.3)	2.2%
X2 + turnover buffer	0.50	+0.19 (p 0.032)	0.22	+0.04 (p 0.274)	3 of 6	+2.8%	-26%	+5.6% (1.5)	1.1%
X3 + volatility targeting	0.44	-0.05 (p 0.545)	0.52	+0.30 (p 0.002) ✓	5 of 6	+5.4%	-16%	+7.4% (2.3)	1.1%

✓ = passes the gate in the holdout. Design Sharpe = average across universes; tests and holdout columns use the six-universe equal-weighted book.

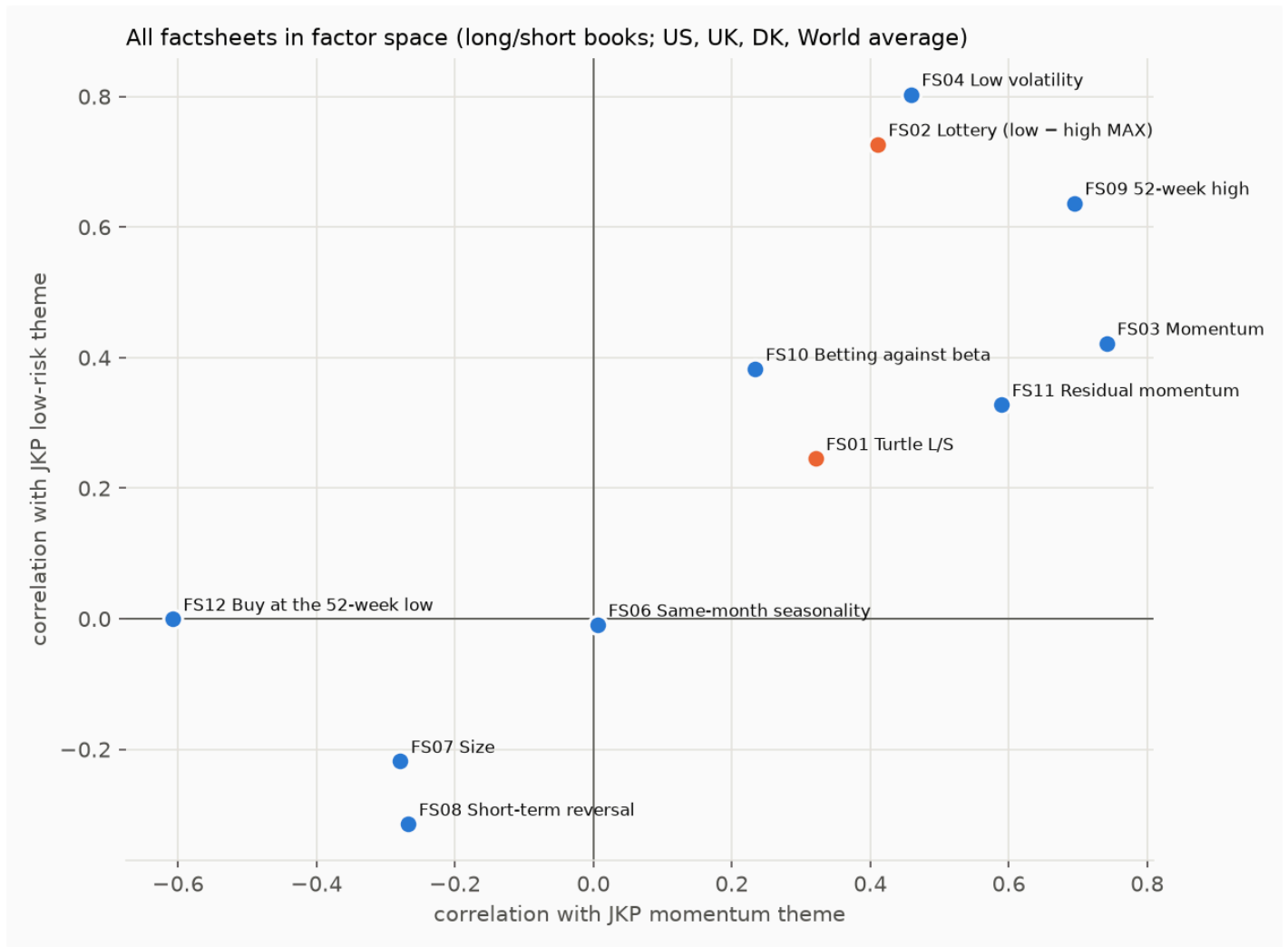


- **X1 beta-neutral legs:** +0.40 design, +0.14 holdout; helps in both windows but does not pass the gate.
- **X2 + turnover buffer:** +0.19 design, +0.04 holdout; helps in both windows but does not pass the gate.
- **X3 + volatility targeting:** -0.05 design, +0.30 holdout; helps in one window and hurts in the other: not reliable.

Where it ends: holdout Sharpe 0.04 → 0.52, return +0.8% → +5.4% a year (six-universe book).

11. Classification: where does it fit?

Dimension	52-week high
Series family	Momentum & trend
Academic style	Anchoring momentum
Signal	Price only (daily total returns)
Cross-section vs time series	Cross-sectional: stocks ranked against each other each month
Horizon and turnover	One month; 83% of the two legs replaced monthly
Market exposure	L/S beta -1.09 to -0.39
Payoff shape	Treynor–Mazuy γ -4.59 to 0.59 (t -3.3 to 0.7); worst 10% of market months +4.3% to +5.5% a month, best 10% -10.1% to -2.5%
Nearest factor theme	momentum (2 of 6 universes), WML (2 of 6 universes)
Nearest library signals (returns)	55-day breakout (6 universes), MIN5 (4 universes), Momentum 12-1 (4 universes)
Economic rationale	Anchoring: traders use the 52-week high as a reference and are reluctant to bid prices above it, so good news is absorbed slowly (George & Hwang 2004).
Publication	George & Hwang 2004
Where it fits	In the momentum cluster of the library; with some information beyond its neighbours.

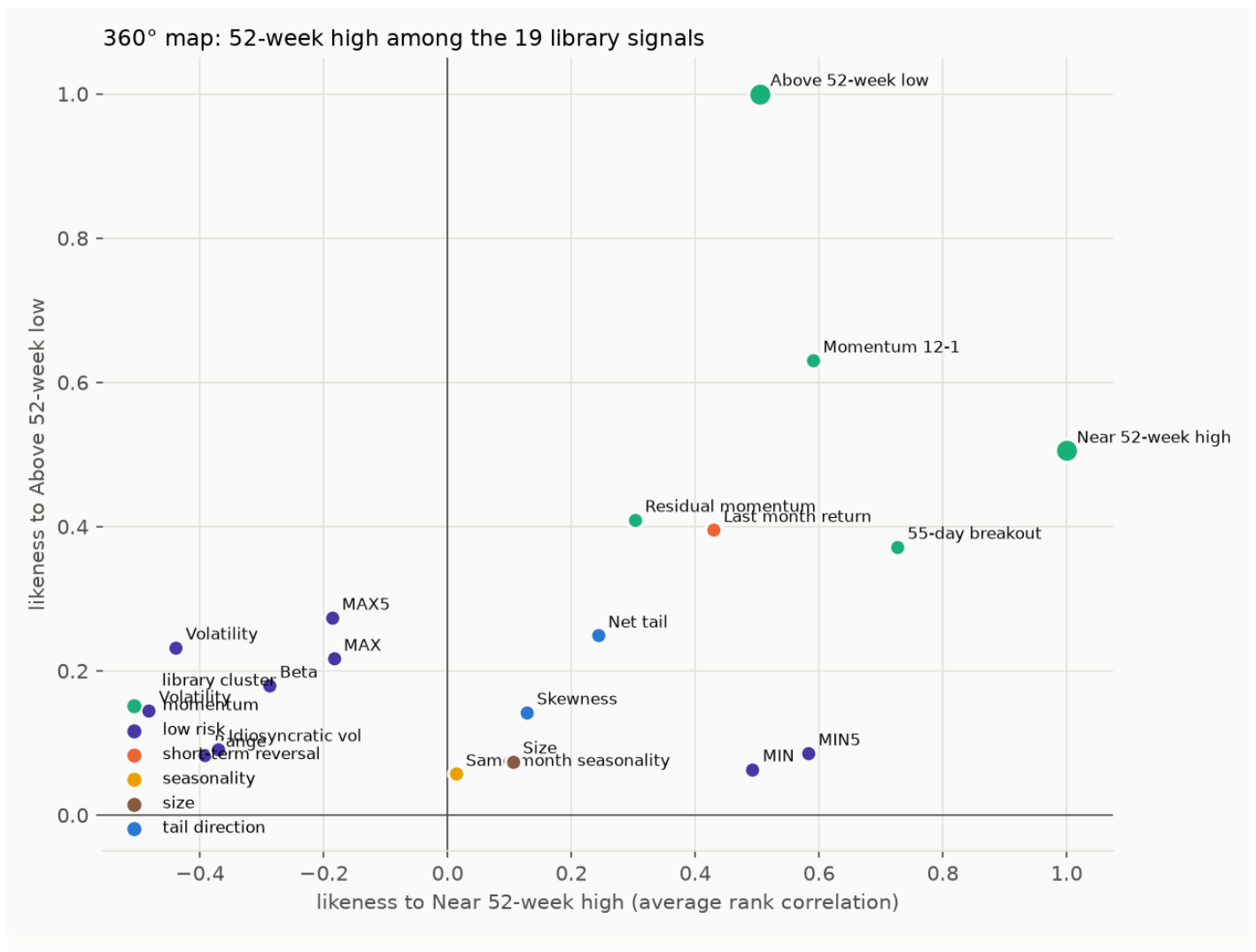


12. 360° view

The signal among the 19 price signals of the shared library (`code/library.py`), on the same universes and months (equal-weighted top minus bottom quantile, gross). Its natural mirror here is **Above 52-week low**.

Signal (top minus bottom, raw sign)	Cluster	Likeness: stocks	Likeness: returns	Raw L/S, avg of 6	CAPM alpha range	Universes alpha t ≥ +2 / ≤ -2
Near 52-week high (this strategy)	momentum	+1.00	+1.00	+3.7%	+9.3% to +22.3%	6 / 0
55-day breakout	momentum	+0.73	+0.83	+1.4%	+4.1% to +13.0%	4 / 0
Momentum 12-1	momentum	+0.59	+0.80	+7.2%	+4.3% to +19.4%	5 / 0
MIN5	low risk	+0.58	+0.79	-0.9%	+2.3% to +13.7%	4 / 0
Volatility	low risk	-0.48	-0.78	+1.1%	-13.7% to -7.9%	0 / 5
MIN (worst day)	low risk	+0.49	+0.74	-0.7%	+3.3% to +11.1%	4 / 0

Signal (top minus bottom, raw sign)	Cluster	Likeness: stocks	Likeness: returns	Raw L/S, avg of 6	CAPM alpha range	Universes alpha $t \geq +2$ / ≤ -2
Volatility (252 days)	low risk	-0.44	-0.74	+2.4%	-12.9% to -4.8%	0 / 3
Range (MAX–MIN)	low risk	-0.39	-0.74	-0.7%	-15.4% to -4.8%	0 / 4
Beta	low risk	-0.29	-0.69	+4.4%	-12.6% to -4.2%	0 / 4
Idiosyncratic vol	low risk	-0.37	-0.62	-3.6%	-15.5% to -4.8%	0 / 4
MAX (best day)	low risk	-0.18	-0.62	+0.1%	-13.3% to -2.0%	0 / 2
MAX5	low risk	-0.19	-0.58	+0.6%	-11.3% to -3.5%	0 / 2
Last month return	short-term reversal	+0.43	+0.52	+1.8%	-3.2% to +11.0%	2 / 0
Residual momentum	momentum	+0.30	+0.46	+6.4%	+8.0% to +12.2%	6 / 0
Above 52-week low (mirror)	momentum	+0.51	+0.46	+7.5%	+5.2% to +13.6%	4 / 0
Size	size	+0.11	+0.36	+0.1%	-3.8% to +9.9%	1 / 0
Net tail (MAX+MIN)	tail direction	+0.24	+0.18	-0.9%	-5.2% to +5.2%	0 / 0
Same-month seasonality	seasonality	+0.01	+0.11	-1.9%	-9.4% to +6.6%	0 / 1
Skewness	tail direction	+0.13	-0.03	-0.4%	-6.7% to +3.9%	0 / 2



Spanning. Regressing the L/S on its five closest library neighbours (by return correlation) and the market:

Universe	Alpha after neighbours / yr	t	R ²	Neighbours with $ t \geq 2$
US	-1.2%	-1.03	0.95	55-day breakout +0.53 (+7.1), Momentum 12-1 +0.41 (+12.5), Volatility -0.23 (-6.3)
EU	+3.8%	2.58	0.93	MIN5 +0.24 (+3.9), 55-day breakout +0.33 (+7.7), Momentum 12-1 +0.45 (+12.2), Volatility -0.22 (-2.9)
UK	+1.0%	0.32	0.88	55-day breakout +0.55 (+6.2), Volatility -0.22 (-2.0), Range -0.24 (-2.4)
DK	+1.8%	0.76	0.74	Momentum 12-1 +0.36 (+4.8), 55-day breakout +0.32 (+4.4), Volatility -0.33 (-4.1), Above 52-week low +0.19 (+2.2)
SCANDI	+3.5%	1.69	0.82	Momentum 12-1 +0.49 (+11.6), 55-day breakout +0.32 (+5.5), Volatility -0.18 (-2.0)
World	-0.9%	-0.92	0.97	55-day breakout +0.48 (+8.9), Momentum 12-1 +0.45 (+14.9), Volatility -0.15 (-3.7)

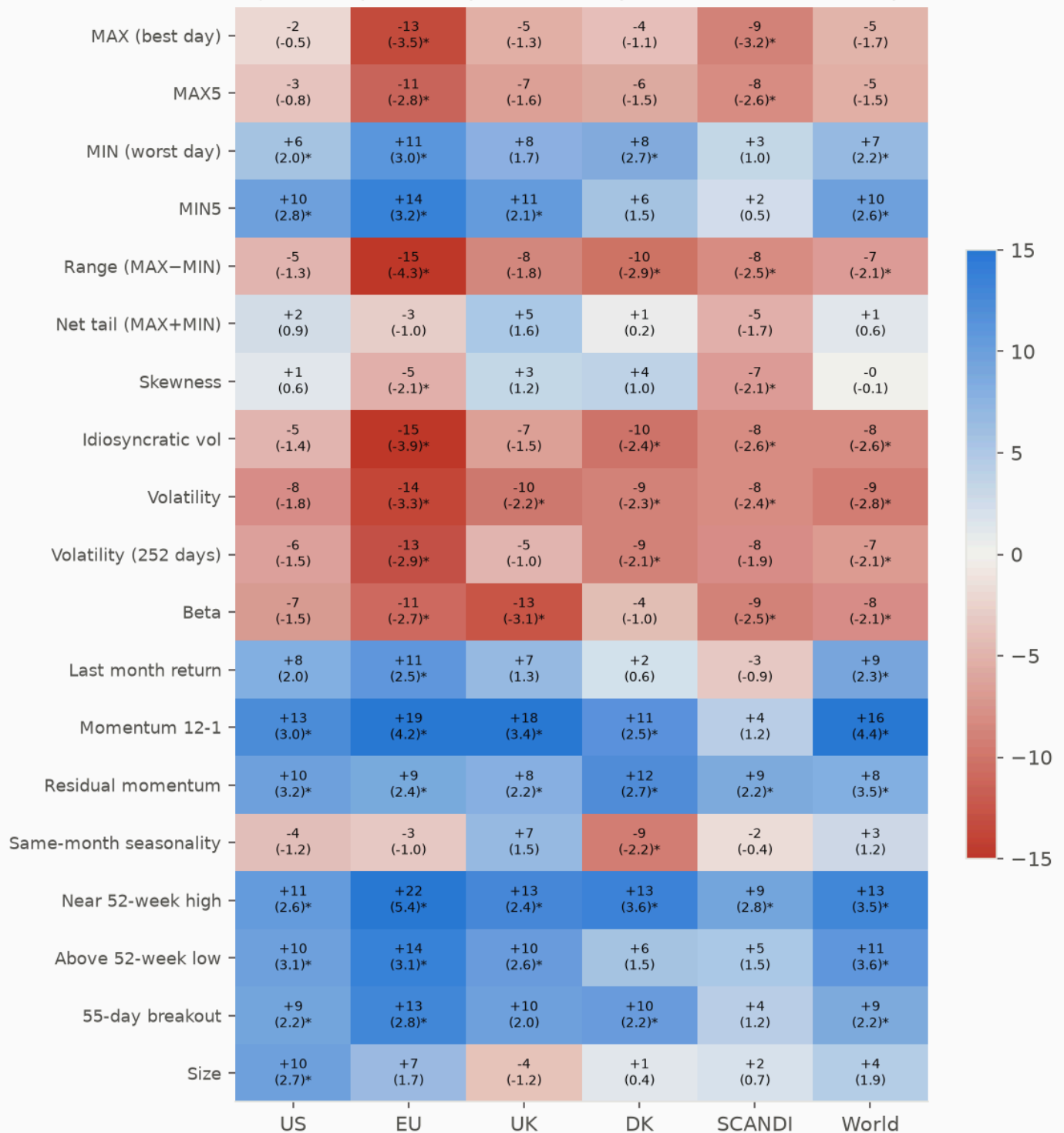
3×3 sort, next-month return a year
(US, EU, UK, SCANDI, World averaged)

low Near 52-week high -	9.4%	11.9%	12.3%
middle -	8.5%	11.2%	14.6%
high Near 52-week high -	9.8%	11.8%	13.5%
	low Above 52-week low	middle	high Above 52-week low

Holding Above 52-week low fixed (down a column), moving from low to high Near 52-week high changes the return by +0.5, -0.1 and +1.2 points a year; holding Near 52-week high fixed (along a row), moving from low to high Above 52-week low changes it by +2.9, +6.1 and +3.7. **Market direction.** Top-minus-bottom return in rising vs falling market months (six-universe average): -15% vs +38% a year. **Horizon:** the gross spread at 1, 2, 3 and 6 months after formation averages +3.7%, +3.5%, +4.0%, +0.3% a year.

The full library, with the CAPM alpha of every signal in every universe:

Signal library: CAPM alpha of each top-minus-bottom sort, % a year (t)



13. Caveats

1. **Residual survivorship.** Point-in-time membership everywhere, but 9–25% of member-quarters outside the US have no price (mostly delisted names); long books are flattered and short books penalised by an unknown amount.
2. **Currency.** Local currency; World sums local-currency returns without conversion.

3. **Equal weighting and flat trading costs.** 10 bp per side for every stock and a size-tiered borrow fee, but no market impact; blue-chip universes only.
4. **Data gap.** We hold for one month; George & Hwang hold for six. A longer holding period would cut turnover.
5. **Not a registered trial.** Descriptive; the fix ladder is counted inside FS03–FS12 (25 trials).

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15. Reproduce

code/data.py → signals.py, signals2.py, library.py → xs.py + analyse_xs.py FS09 → build_xs.py FS09.
Metrics use Project1 InvestmentLibrary. Licensed and cached prices are not included.